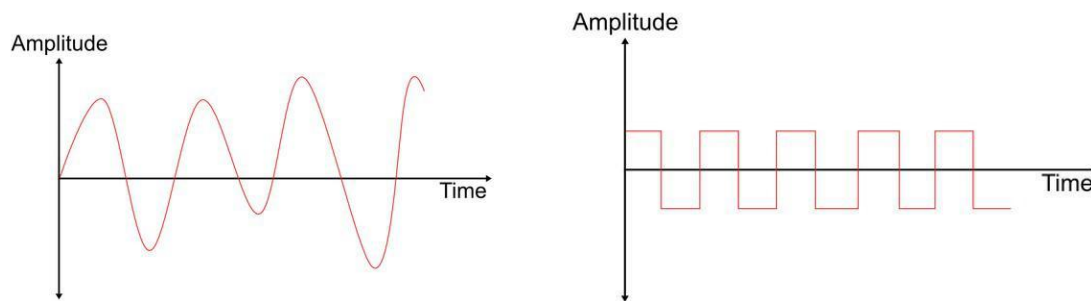


Analog Data:

1. Analog data represents information in a continuous form.
2. It can take infinite values within a given range.
3. Examples include human voice, music and temperature.
4. An analog clock shows time through the continuous movement of its hands.
5. Analog data such as sound creates continuous waves in the air, which can be captured by a microphone and converted into analog or digital signals.

Digital Data:

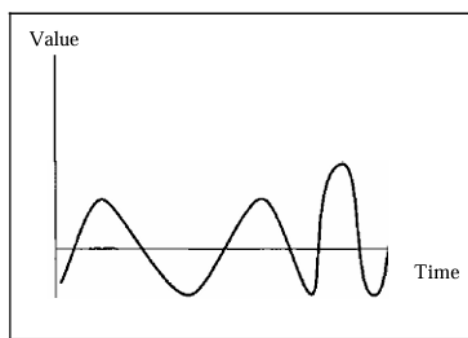
1. Digital data represents information using discrete values.
2. It has a limited number of states, usually represented as 0s and 1s.
3. Computers store and process data in digital form.
4. A digital clock changes time suddenly, such as from 8:05 to 8:06.
5. Digital data can be transmitted as digital signals or converted into analog signals for communication over a medium.

**Analog Signal:**

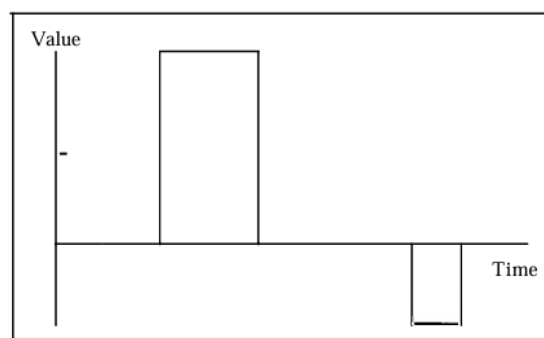
1. An analog signal represents data using continuous waveforms.
2. It has infinitely many levels of intensity over a period of time.
3. The signal changes smoothly from one value to another without sudden jumps.
4. Examples include sound waves, radio waves and telephone signals.
5. On a graph, an analog signal appears as a smooth, continuous curve with time on the horizontal axis and amplitude on the vertical axis.

Digital Signal:

1. A digital signal represents data using discrete values.
2. It has a limited number of defined levels, usually 0 and 1.
3. The signal changes suddenly from one value to another.
4. Digital signals are commonly used in computers and digital communication systems.
5. On a graph, a digital signal appears as a series of square waves or pulses.



a. Analog signal



b. Digital signal

Periodic Analog Signals: Periodic analog signals can be classified as simple or composite. A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.

Sine Wave: A sine wave is a smooth, continuous waveform used to represent an analog signal. It shows how a signal varies with time in a regular and repeating manner.

1. It is characterized by a smooth oscillation above and below a central axis.

2. A sine wave is defined by three main parameters:

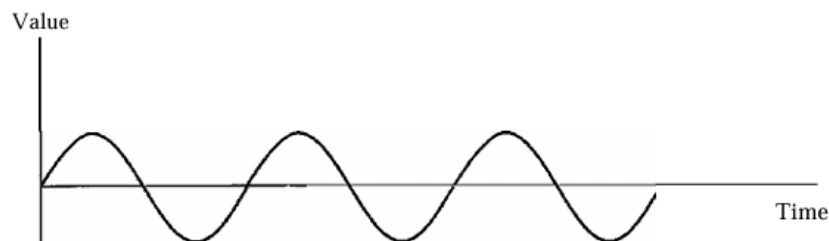
- **Amplitude:** Maximum strength or height of the wave
- **Frequency:** Number of cycles completed per second. Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

- **Phase:** Position of the wave at a given time

3. Sine waves are commonly used to represent sound waves, radio signals and electrical signals.

4. It is the basic waveform used in signal analysis because complex signals can be formed by combining sine waves.



Peak Amplitude:

1. Peak amplitude is the maximum value or strength of a signal measured from the zero (reference) level.

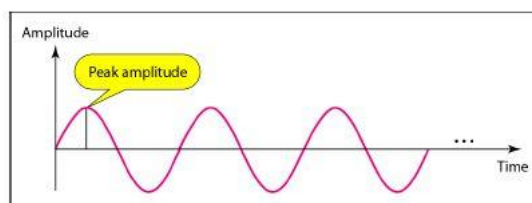
2. It shows how high or low a signal goes during one cycle.

3. Peak amplitude represents the energy or power of the signal.

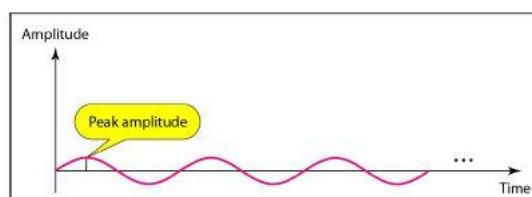
4. In analog signals, a higher peak amplitude means a stronger signal.

5. Peak amplitude is usually measured in units like volts or amperes.

Two signals with the same phase and frequency, but different amplitudes



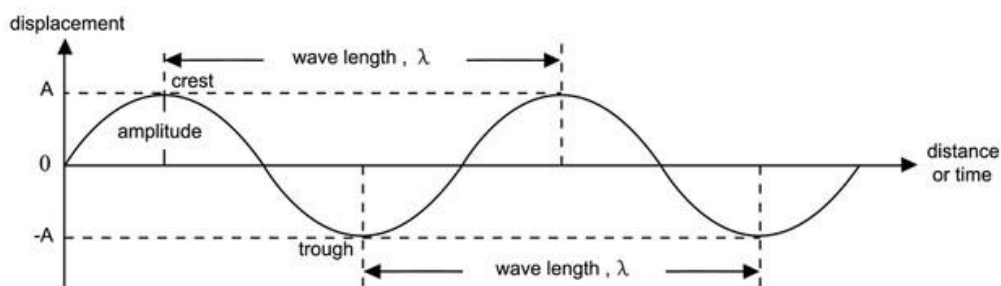
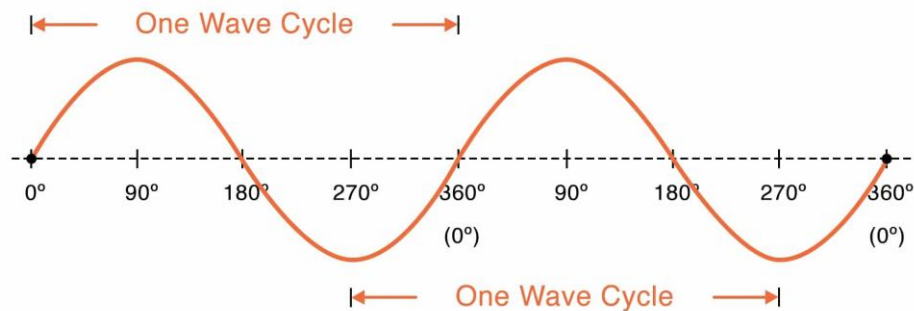
a. A signal with high peak amplitude



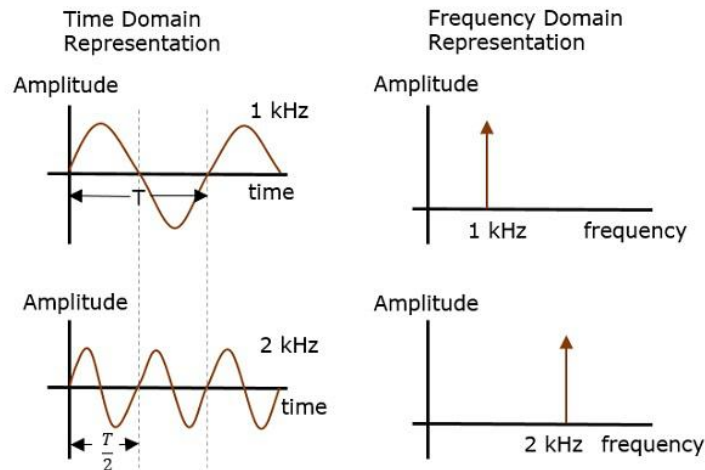
b. A signal with low peak amplitude

Wavelength:

1. Wavelength is the distance a signal travels in one complete period of a sine wave.
2. It is a characteristic of a signal traveling through a transmission medium.
3. Wavelength depends on both the frequency of the signal and the propagation speed of the medium.
4. The frequency of a signal remains constant, but the wavelength changes when the medium changes.
5. Wavelength is used in data communication, especially to describe light transmission in optical fibers.
6. Wavelength is represented by λ (lambda) and is calculated as: $\lambda = \frac{c}{f}$ where c is propagation speed and f is frequency.
7. The speed of light is about 3×10^8 m/s in vacuum, and it is lower in air and cables.
8. Wavelength is usually measured in micrometre's (μm).

**parts of a wave****Phase:**

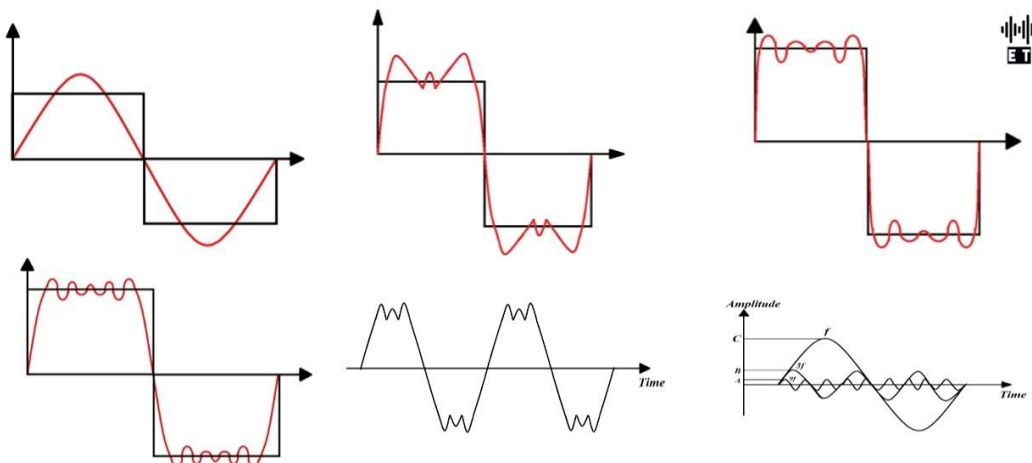
1. Phase describes the position of a signal's waveform at a specific point in time.
2. It indicates how much a wave is shifted relative to a reference point or another wave.
3. Phase is usually measured in degrees ($^\circ$) or radians.
4. Two signals with the same frequency but different phase do not reach their peaks at the same time.
5. Phase is important in signal synchronization and data communication.

Time and Frequency Domains:**Time Domain Representation:**

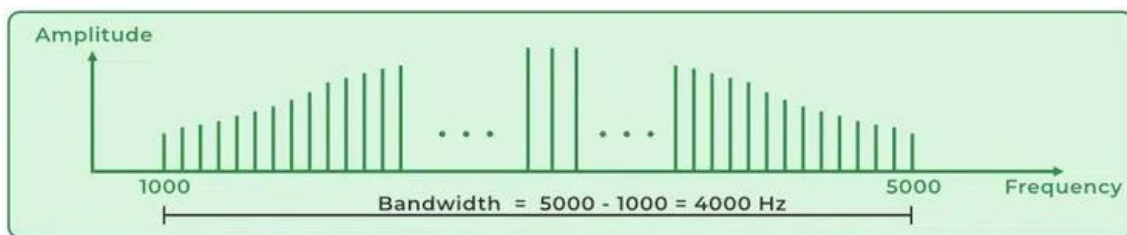
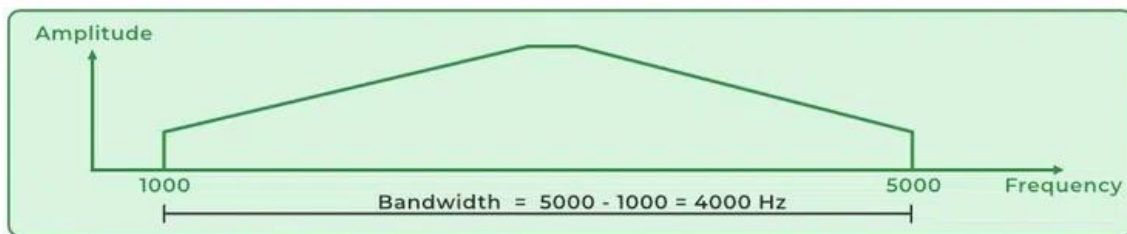
1. In the time domain, a signal is shown as how its amplitude changes with time.
2. The horizontal axis represents time, and the vertical axis represents signal strength (amplitude).
3. It shows the actual shape of the signal as it varies over time.
4. Time domain representation is easy to understand and is commonly used to observe signal behavior, timing and duration.
5. Most real-life signals like speech and data signals are first observed in the time domain.

Frequency Domain Representation:

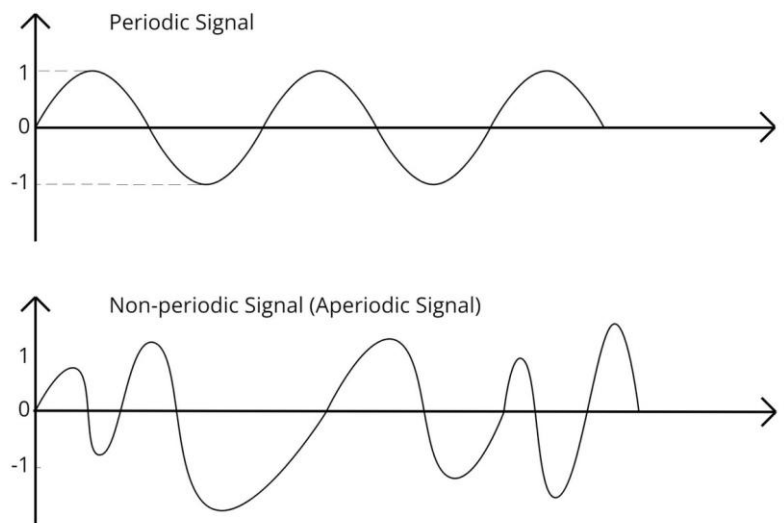
1. In the frequency domain, a signal is represented by its frequency components instead of time.
2. The horizontal axis represents frequency, and the vertical axis represents amplitude of each frequency.
3. It shows which frequencies are present in the signal and how strong each one is.
4. Frequency domain representation is based on Fourier analysis, which states that a signal can be expressed as a combination of sine waves.
5. It is useful for finding the bandwidth of a signal and is important in signal analysis and communication system design.

Composite Signals:

1. A composite signal is formed by combining many simple sine waves. A single sine wave is useful for carrying energy or simple alerts, but it cannot carry complex information like voice or data. Therefore, data communication uses composite signals.
2. Most real-life signals such as speech, music and video are composite signals because they contain many frequencies.
3. According to Fourier analysis, any composite signal can be represented as a combination of sine waves with different frequencies, amplitudes and phases.
3. Composite signals can be periodic or nonperiodic. Periodic composite signals have discrete frequencies that are multiples of a fundamental frequency, while nonperiodic composite signals have continuous frequencies.
4. The range of frequencies present in a composite signal is called its bandwidth, which determines the data-carrying capacity of the signal.

Bandwidth:**a. Bandwidth of a Periodic Signal****b. Bandwidth of a Non-Periodic Signal**

1. Bandwidth is the range of frequencies that a signal occupies or that a communication channel can transmit.
2. It is defined as the difference between the highest and lowest frequencies present in a signal.
3. Bandwidth determines the data-carrying capacity of a communication system.
4. A higher bandwidth allows more data to be transmitted in less time, while lower bandwidth limits transmission speed.
5. Bandwidth is usually measured in hertz (Hz) and is very important for transmitting voice, video and data signals efficiently.

Periodic and Non-periodic Signal:**PERIODIC SIGNAL AND APERIODIC SIGNAL****Periodic Signals:**

1. A periodic signal is a signal that repeats itself at regular time intervals, called the period.
2. Each cycle of the signal is identical in shape and duration.
3. Periodic signals are characterized by amplitude, frequency, period and phase.
4. Simple sine waves and signals used in power transmission are examples of periodic signals.
5. According to the Fourier series, a periodic signal can be represented as a sum of sine waves with discrete frequencies.

Nonperiodic Signals:

1. A nonperiodic signal is a signal that does not repeat its pattern over time.
2. The waveform changes continuously and has no fixed period.
3. Nonperiodic signals are usually described using bit rate instead of frequency.
4. Signals such as speech, music and data transmission are nonperiodic in nature.
5. According to Fourier analysis, a nonperiodic signal consists of infinitely many sine waves with continuous frequencies.

Digital Signal:

1. In a digital signal, data is represented using discrete voltage levels. For example, binary 1 is represented by a positive voltage and binary 0 by zero voltage.
2. A digital signal can have more than two levels. When more levels are used, more than one bit can be sent per signal level, which increases the data rate. If a digital signal has L levels, then the number of bits represented per level is: **Bits per level = $\log_2 L$**
3. Digital signals are usually nonperiodic, so frequency and period are not used to describe them. Instead, bit rate is used to represent the speed of data transmission.

Bit Rate: Bit rate is the speed at which data is transferred in a communication system. It is measured in bits per second (bps) and may also be expressed as Kbps, Mbps, or Gbps for higher speeds. Bit rate is also called data rate and can be calculated from system bandwidth by dividing the bandwidth by 8.

Bit Length: Bit length is the distance occupied by one bit on a transmission medium. It depends on the propagation speed of the signal and the duration of one bit. Bit length is calculated by multiplying the propagation speed by the bit duration.

Transmission of Data Signals: We can transmit a digital signal by using one of two different approaches, baseband transmission or broadband transmission.

1. Baseband transmission is the method of sending a digital signal directly over a communication channel without converting it into an analog signal.
2. It requires a low-pass channel, meaning the channel bandwidth starts from zero frequency and supports only one channel.
3. It is used when a dedicated medium is available, where the entire bandwidth is used for a single communication.
4. Examples include a direct cable connection between two computers or a bus network where only two stations communicate at a time, using the full bandwidth.

Baseband transmission

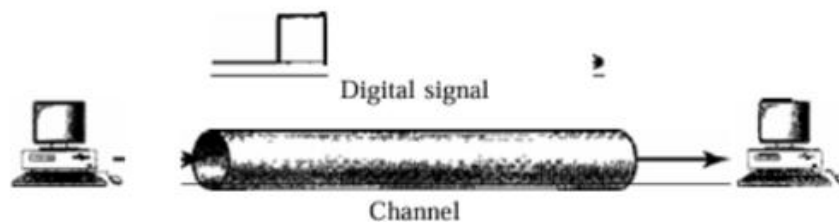
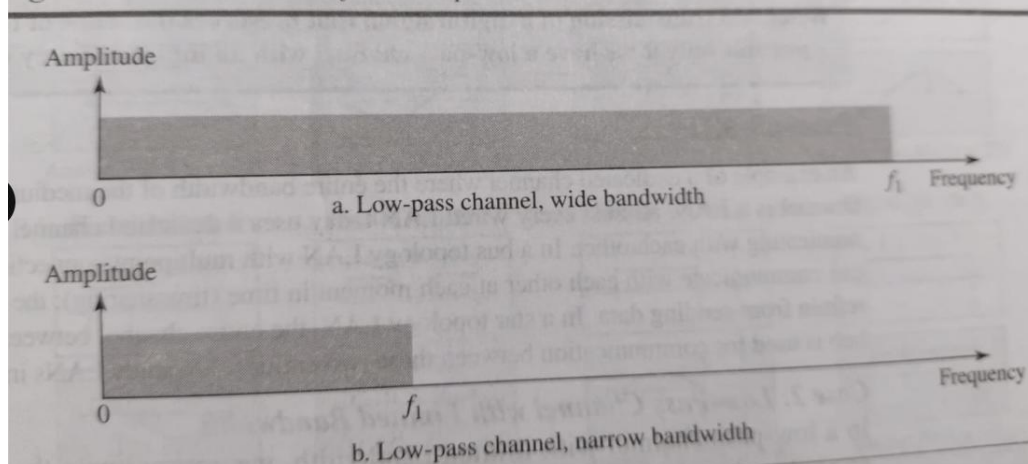


Figure 3.20 Bandwidths of two low-pass channels

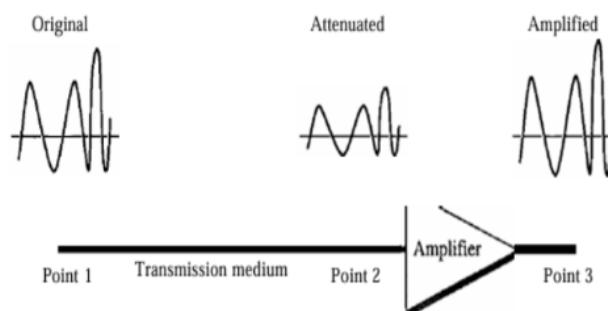


Transmission Impairment: When a signal travels through a transmission medium, the medium's imperfections cause the received signal to differ from the transmitted signal. The three main causes of transmission impairment are attenuation, distortion, and noise.

1.Attenuation:

- Attenuation refers to the loss of signal energy as it travels through a transmission medium.
- This energy is lost due to the resistance of the medium, and some of it is converted into heat, which is why wires may become warm.
- To overcome this loss and restore signal strength, amplifiers are used.

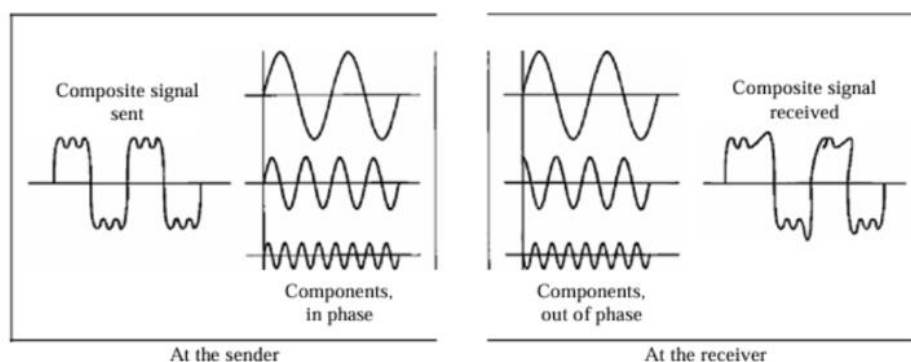
Figure 3.26 Attenuation



2.Distortion:

- Distortion occurs when a signal changes its shape or form during transmission.
- It mainly affects composite signals, which are made up of different frequencies.
- Different frequency components travel at different speeds, causing delays and phase changes.
- As a result, the received signal shape is different from the transmitted signal.

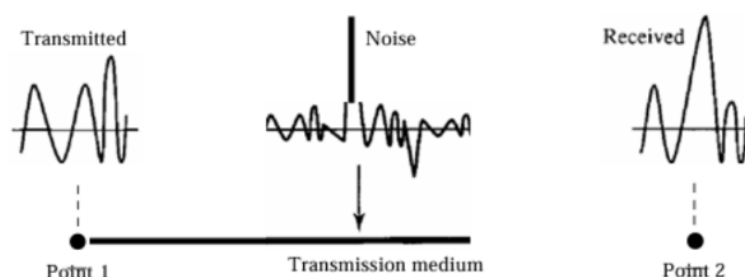
Figure 3.28 Distortion



3.Noise:

- Noise is an unwanted disturbance that affects a signal during transmission and causes signal impairment.
- There are different types of noise:
 - ❖ **Thermal noise:** Caused by random motion of electrons in a wire.
 - ❖ **Induced noise:** Comes from external sources like motors and electrical appliances.
 - ❖ **Crosstalk:** Occurs when one wire affects another nearby wire.
 - ❖ **Impulse noise:** Sudden high-energy spikes caused by lightning or power lines.
- Noise can corrupt or distort the signal, reducing the quality of communication.

Figure 3.29 Noise



Performance: Network performance refers to how efficiently and effectively a network operates. It is often measured using Quality of Service (QoS), which evaluates the overall performance of the network.

The main factors used to measure performance are:

1. **Bandwidth:** The maximum amount of data that can be transmitted over a network in a given time.
2. **Throughput:** The actual amount of data successfully transmitted over the network.
3. **Latency (Delay):** The time it takes for data to travel from the sender to the receiver.

These parameters help in determining how good and reliable a network is.

Bandwidth: Bandwidth is an important factor used to measure network performance. The term bandwidth is used in two different ways:

1.Bandwidth in Hertz (Hz): It refers to the range of frequencies that a signal contains or that a channel can transmit. For example, a telephone line has a bandwidth of 4 kHz.

2.Bandwidth in Bits per Second (bps): It refers to the maximum number of bits that a channel, link, or network can transmit per second. For example, Fast Ethernet has a bandwidth of 100 Mbps.

3.Relationship: There is a relationship between bandwidth in hertz and bandwidth in bits per second. Generally, an increase in frequency bandwidth leads to an increase in data transmission rate. This relationship depends on whether the transmission is baseband or modulated.

Throughput:

1. Throughput is the actual rate at which data is successfully transmitted through a network.
2. It is different from bandwidth; while bandwidth shows the maximum capacity of a link, throughput shows the real data transfer rate.
3. Throughput is usually less than bandwidth due to limitations such as device performance, network congestion, or other factors.
4. For example, a link may have a bandwidth of 1 Mbps, but if connected devices support only 200 kbps, then the actual throughput is 200 kbps.

Latency (Delay):

1. Latency is the total time taken for a message to reach the destination from the moment the first bit is sent.
2. It consists of four components:
 - ❖ Propagation time
 - ❖ Transmission time
 - ❖ Queuing time
 - ❖ Processing delay
3. The total latency is calculated as:
$$\text{Latency} = \text{Propagation time} + \text{Transmission time} + \text{Queuing time} + \text{Processing delay}$$
4. Propagation time is the time required for a bit to travel from source to destination and is calculated as:
$$\text{Propagation time} = \text{Distance} / \text{Propagation speed}$$